D64.

* In closed-loop settings, the measurements are dependent on the MHE estimates!

$$\hat{x}_{k-1}, \hat{d}_{k-1} \rightarrow u_{k-1}(\hat{x}_{k-1}, \hat{d}_{k-1}, x^{ref}_{k-1}) \rightarrow x_k \rightarrow y_k = h(x_k) + v \rightarrow \text{MHE update}$$

Added into my PhD Thesis.

30 - Oct. 2023. · Intuitive Explanation of Online Learning NeuroMHE.

~~Sup~~ Supervised learning of NeuroMHE is intuitive ~~and~~, straightforward, and effective, as it directly measures the deviation of the MHE estimates away from their ground truth values, which is the core target of state estimation.

$$\hat{x} \xrightarrow{\text{MHE}} x \text{ (ground truth)}$$

Now a question arises, why does RL of NeuroMHE from the tracking errors work, at least via the simulations using the synthetic dataset? Minimization the error between the MHE estimates and the corresponding reference states is seemingly counter-intuitive for state estimation.

$$\hat{x} \xrightarrow[R\uparrow\ Q\downarrow]{\text{MHE}} x \xrightarrow[\hat{d}\rightarrow d]{\text{Control}} x^{ref} \text{ (reference state)}$$

Below is an explanation of why such an tracking error-based ~~loss~~ learning works. (Note that it works only under closed-loop robust control). At the core of the explanation is to show that minimizing the tracking error functions in the same manner as minimizing the estimation error does! Let's show it by contradiction.

- Our expectation for learning: ~~An~~ Effective learning should lead to an increase in the weights related to the measurement error while reducing the weights related to the process noises. $R\uparrow$ $Q\downarrow$ (So, we are allowed to change the process noises for tracking the relatively accurate measurement. In this way, the estimated disturbances should converge toward their ground truth values)

- Opposing results: After training, the weights related to the measurement error decreases but there is a significant increase in the weights for the process noises. $R\downarrow$ $Q\uparrow$ So, we have very limited confidence in the measurement and want to suppress the process noises significantly. (The optimal disturbances will (are more likely to) be zeros.

We are going to show that the fulfillment of the control objective and the opposing results cannot co-exist! (Note that the feedback controller takes as input the current estimate, including the disturbance estimate, from NeuroMHE)

$$\min_{X,W} J = \frac{1}{2}\|x_0 - \hat{x}_0\|_P + \sum_{k=0}^{N} \frac{1}{2}\|y_k - h(x_k)\|^2_R + \sum_{k=0}^{N-1} \frac{1}{2}\|w_k\|^2_Q$$

* Initial guess of $\hat{d}_k$ is zero!

$$\text{s.t.} \begin{cases} \hat{x}^q_{k+1} = \cancel{f(x_k)}\ \hat{x}^q_k + \Delta t \cdot \big(f(\hat{x}^q_k) + \underbrace{[-f(\hat{x}^q_k) + \dot{x}^{ref}_k - \hat{d}_k]}_{u := \text{numerical value input for MHE}} + \hat{d}_k\big) \\ \hat{d}_{k+1} = \hat{d}_k + \Delta t \cdot \hat{w}_k \end{cases}$$

(robust feedback controller taking the disturbance estimate as the feedforward term for disturbance compensation)

$x := [x^q, d]$

In the opposing results, $R\downarrow$ $Q\uparrow$ result in nearly zero estimate of the disturbance $d_k \approx 0$. If R is very small, the resulting MHE and the controller are decoupled from the robotic system, as the MHE estimator does not need real measurement sampled from the real system! This means that the state estimates from MHE ~~[illegible]~~ need not be related to the measurement / states of the real system, much less the reference states. This is because, in theory, there is no any forms of constraint on the system states in the cost. Satisfying the dynamics equality constraint does not ensure that the state estimates can track the reference states! → The opposing results cannot occur!

when $\hat{d} \to d$, then $x \to x^{ref}$, leading to $y \to y^{ref}$, and this in turn encourages $\hat{x}$ to become x^{ref} 55.

else
when $\hat{d} \not\to d$, then $x \not\to x^{ref}$, leading to $y \not\to y^{ref}$, so $\hat{x}^{MHE}$ deviates from x^{ref}.

In the MHE cost, $\|y_k - h(x_k)\|$ is a cost that can guide x_k towards x^{ref} through $y \to y^{ref}$.

30-Dec-2023.

continued discussion of the tracking error-based RL of NeuroMHE.

2-May-2026

For an MHE KKT system:

$F(z, \lambda; \theta, Y, U) = 0$.

then exact closed-loop differentiation gives:

$$F_{z,\lambda} \frac{d(z,\lambda)}{d\theta} = -\left(F_\theta + F_Y \frac{dY}{d\theta} + F_U \frac{dU}{d\theta}\right)$$

considered in the NeuroMHE paper. (for the frozen-date MHE layer)

$$\min_\theta \; Loss\left(\hat{x}^q(\theta), x^{q,ref}\right)$$

s.t. $\hat{x}^q(\theta), \hat{d}(\theta) = MHE(\underline{y}, u, \theta)$ $\longrightarrow$ MHE is coupled with the true system by the measurement y sampled from the system.

$u = U(\hat{x}^q, \hat{d}, x^{q,ref})$

$y = Model(x, u)$ $\longrightarrow$

If $\hat{x}^q$ deviates from y significantly as a result of a very small R penalizing the measurement error, MHE is then decoupled from the robot system and the stability of the MHE estimates is at risk.

$$\hat{d} \to d \iff \hat{x}^q \to x^{q,ref}$$

We have shown in the left page that $\hat{x}^q \to x^{q,ref} \Rightarrow \hat{d} \not\to d$ is impossible

$\hat{x}^q \to x^{q,ref}$ is possible if and only if $\hat{d} \to d$.

So, in other words, (intuitively) $\hat{d} \to d$ is a major (almost the sole) factor that contributes to the improved robust tracking performance. Minimizing this loss can ~~using~~ encourage ~~to~~ pushing the ~~MHE~~ disturbance estimate $\hat{d}$ toward the ground truth.

* 2-May-2026. For the exact closed-loop objective, we need to consider the partial derivative of the measurement y and the control u w.r.t. θ in MHE!

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